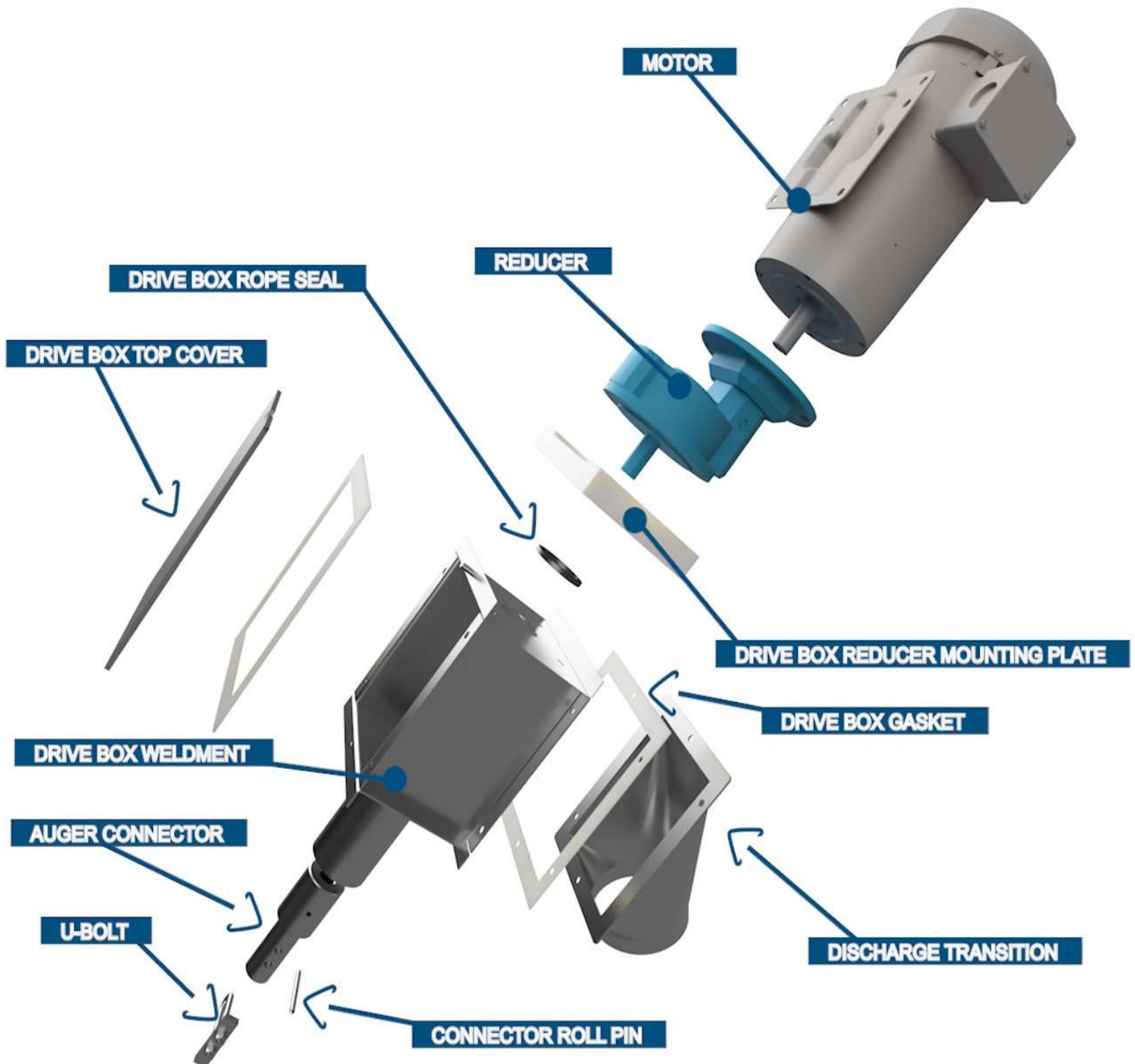


5.0-Helix Introduction

Hapman Helix™ Conveyors are the most efficient, easy to operate, and maintainable conveyors on the market. With a single moving part in the standard Helix™, there will be minimal maintenance required. We designed our conveyors to provide customers with an economical conveying method that can be used for most dry material handling applications.

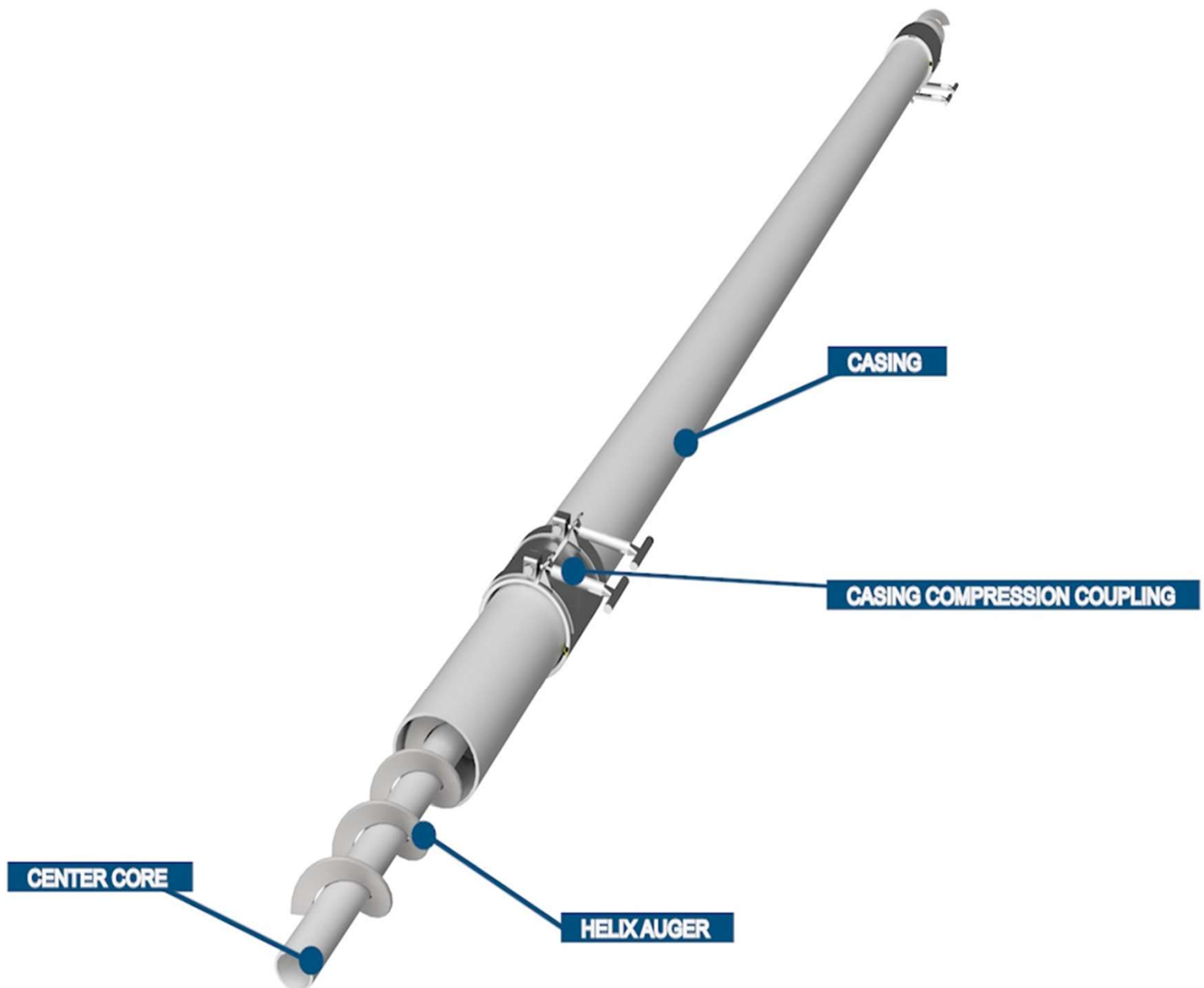


6.0 Helix 250-300 Series Major Components: Drive Assembly



6.0 Helix 250-300 Series Major Components: Auger Assembly

There are four main components that will be installed with the Helix™ Auger Assembly. They are the casing, compression couplings, auger and center core*.



** Not all applications require a center core so please reference the bill of materials provided with your equipment.*

6.0 Helix 250-300 Series Major Components: Auger Assembly

Stepped Gasket Compression Coupling

The stepped gasket design on the compression couplings allows you to secure the larger diameter inlets on the drive boxes and hopper boots to the smaller diameter poly Helix™ casing. A secondary gasket is used on one side to “step” down the ID of the coupling to match the poly Helix casing. While the straight gasket on the opposite side of the coupling maintains a larger ID to easily fit to the drive box inlet and hopper boot.



Straight Gasket:
This side mounts
onto the metal
drive box and
hopper.

Stepped Gasket:
This side goes onto
the poly casing.

6.0 Helix 250-300 Series Major Components: Auger Assembly

Straight Gasket Compression Coupling

The straight gasket design on the compression couplings are utilized when a split poly casing design is implemented. It allows you to join two pieces of poly casing together that are the same diameter.

NOTICE

Straight couplings are also utilized on units with a steel casing. See Appendix G for installation of a steel casing.

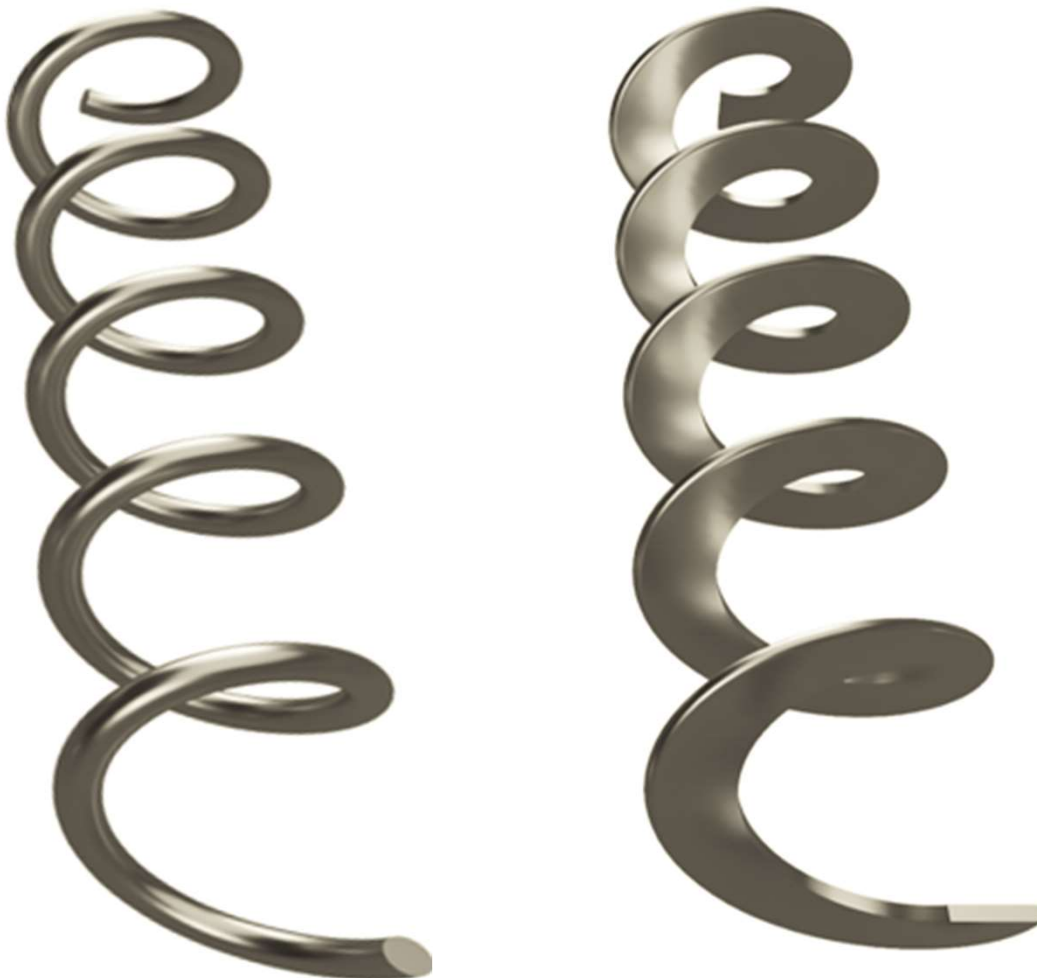


6.0 Helix 250-300 Series Major Components: Auger Assembly

There are four main types of Helix™ augers that Hapman supplies with its units. Each one is uniquely suited for different applications and materials that are introduced into the conveyor.

Round Wire - Used with fine granular material and dense granular materials such as salt, sugar. It is also utilized for larger particle materials where the integrity of the particle must remain intact like coffee beans or dried corn kernels. The round auger provides some “slip” within the material reducing the load on the conveyor.

Flat Wire - used for lighter free flowing fine powders such as flour, starch, and talc. It provides the highest conveyor capacity.



6.0 Helix 250-300 Series Major Components: Auger Assembly

There are four main types of Helix™ augers that Hapman supplies with its units. Each one is uniquely suited for different applications and materials that are introduced to the conveyor.

Beveled Edge - is utilized with materials that like to build up or stick to the interior of the Helix™ casing. It has a mitered cut on the trailing edge of the screw. The reduced surface area of the outer edge along with the beveled trailing edge reduces material build up within the casing.

HD Square Bar - used for materials with an extremely high bulk density or highly abrasive materials. This auger requires the use of a steel Helix™ casing.

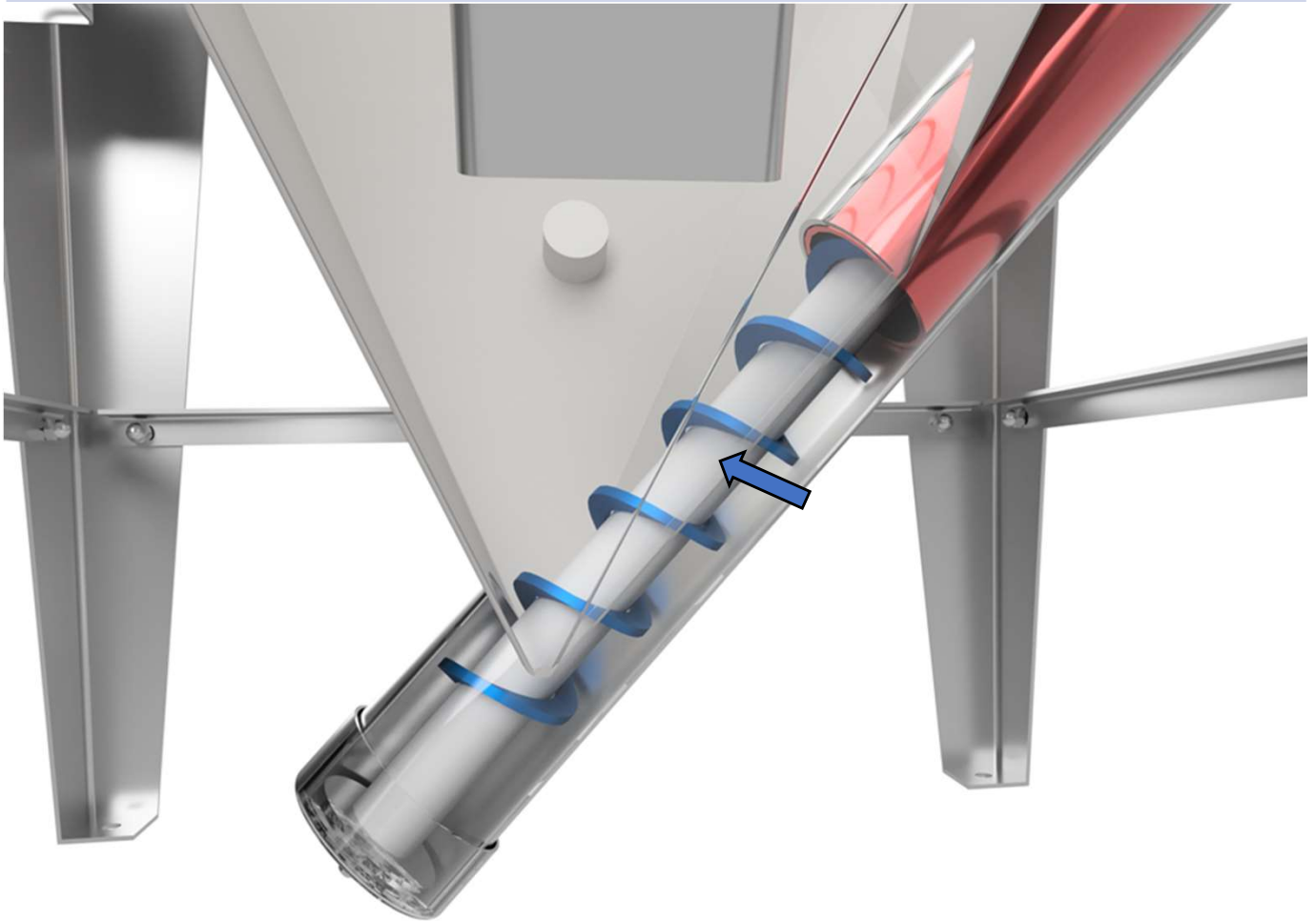


6.0 Helix 250-300 Series Major Components: Auger Assembly

Helix Center Core

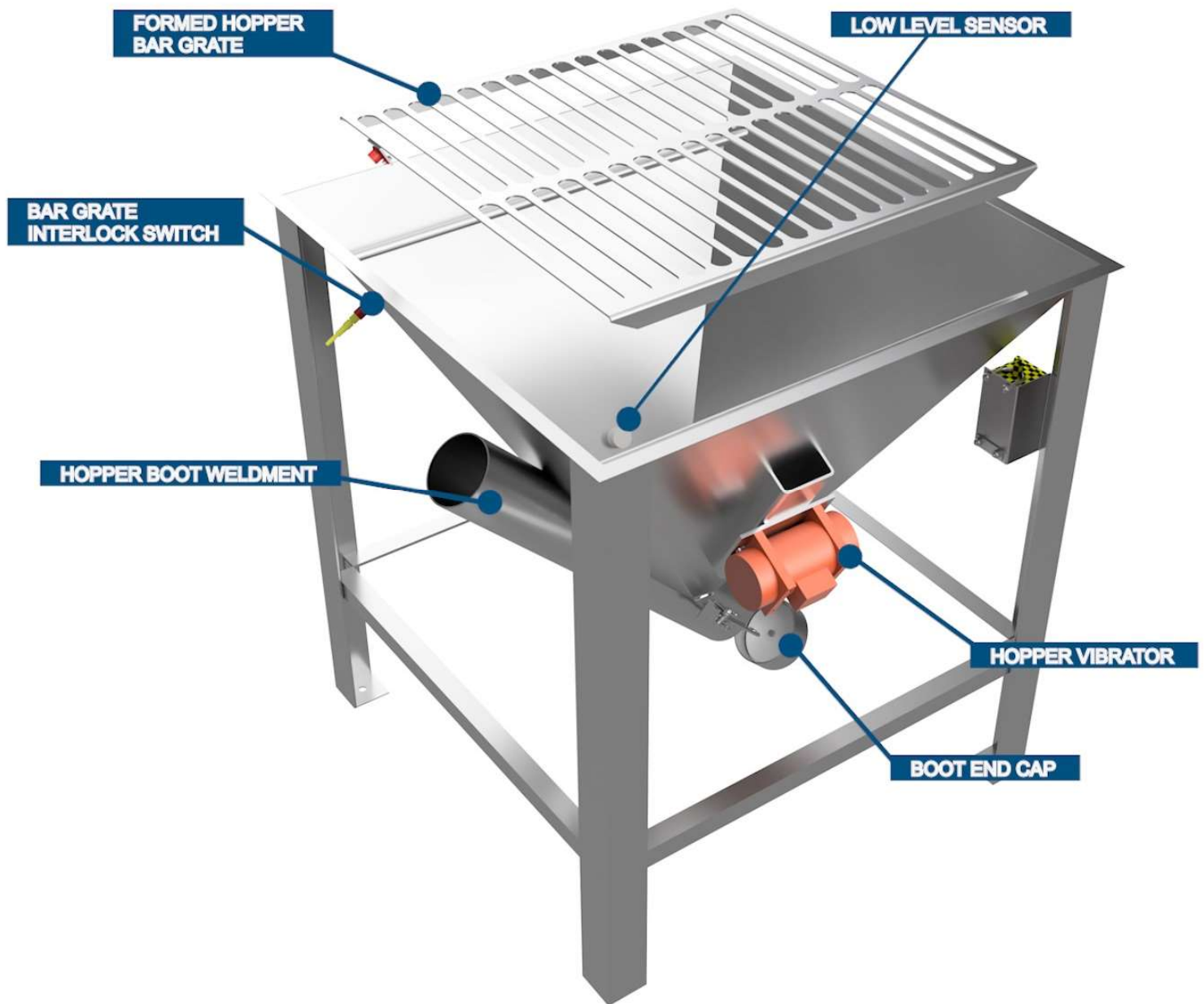
The Helix™ center core is a removable polyethylene tube floating within the auger interior and is effective for very fine materials that tend to aerate and become fluid. The core fills the auger interior and reduces the flow back of the fluid material.

**Not all applications require a center core so please reference the bill of materials provided with your equipment.*



6.0 Helix 250-300 Series Major Components: Hopper Assemblies

Hopper with removable bar grate and safety interlock sensor



6.0 Helix 250-300 Series Major Components: Hopper Assemblies

Hopper with hinged bar grate and mechanically interlocked limit switch.

On this style hopper the “Bar Grate Limit Switch” is referred to as a “Mechanical Interlock” on the electrical prints and will need to be wired per the prints for the unit to run. Reference appendix F in this manual for instructions

